# Week 8 | Solve With Instructors

Q1) The following table represents the datasets of 10 objects, with attributes Color, Type, Origin, and the "class", whether the customer who bought was satisfied or not:

| Sr. No. | Color | Type | Origin | Satisfied? |
|---|---|---|---|---|
| 1 | Red | Casual | Domestic | Yes |
| 2 | Red | Casual | Domestic | No |
| 3 | Red | Casual | Domestic | Yes |
| 4 | Yellow | Casual | Domestic | No |
| 5 | Yellow | Casual | Imported | Yes |
| 5 | Yellow | Casual | Imported | Yes |
| 6 | Yellow | Formal | Imported | No |
| 7 | Yellow | Formal | Imported | Yes |
| 8 | Yellow | Formal | Domestic | No |
| 9 | Red | Formal | Imported | No |
| 10 | Red | Casual | Imported | Yes |

We want to classify a new object with the following new properties: Color=Red, Origin=Domestic and Type=Formal. Use Naive Bayes algorithm and tell the class of this test instance.

Q2) $x = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix}^T$ is a feature vector and $y$ represents the label in binary classification problem, Which of the following equations represents the 'naive' assumption used in the Naive Bayes algorithm.

A) $P(x, y) = P(y) \cdot P(x \mid y)$

B) $P(x) = P(x_1) \cdot P(x_2) \cdot P(x_3)$

C) $P(y \mid x) = \dfrac{P(y) \cdot P(x \mid y)}{P(x)}$

D) $P(x \mid y) = P(x_1 \mid y) \cdot P(x_2 \mid y) \cdot P(x_3 \mid y)$

Q3) A Gaussian Naive Bayes model is trained for a multi-class classification problem that has 10 features and 3 classes. Find the total number of parameters that have to be estimated for this model. Consider each parameter to be a scalar value. In other words, if we decide to store all the parameters in a Python list, with each element of the list being a float value corresponding to a single parameter, what is the size of this list? Ignore the priors in the calculation. Only focus on the parameters of the class conditional densities.

Q4) You have trained a Naive Bayes model on a dataset with two classes (positive and negative sentiment). For a new instance, the model provides the following conditional probabilities:

$$P\big(X_1 = x_1 \mid \text{Positive}\big) = 0.6, P\big(X_2 = x_2 \mid \text{Positive}\big) = 0.5$$
$$P\big(X_1 = x_1 \mid \text{Negative}\big) = 0.2, P\big(X_1 = x_1 \mid \text{Negative}\big) = 0.3$$

The prior probabilities of $P\big(\text{Postive}\big)$ is $0.4$. What is the posterior probability for each class?

A Gaussian naive Bayes model is trained on a given dataset. For an unseen data point x, the following two values are calculated as

$$P(x|y=0) = 0.4 \; P(x|y=1) = 0.6$$

What will be the predicted label for $x$ ?

(a) 0

(b) 1

(c) Insufficient information to make prediction

. You know the distribution $P(X, y)$ for a given dataset $X, y$. Can you always find the distribution $P(y|X)$ for the same dataset $X, y$

(a) Yes

(b) No

. Suppose you have a three-class classification problem where class label $y \in 0, 1, 2$ and each training example $x_i$ has binary features $f_1, f_2, f_3 \in 0, 1$. How many parameters do you need to know to classify an example using the Naive Bayes classifier ?

Consider a naive Bayes model is trained on the following data matrix $X$ of shape $(d, n)$ and corresponding label vector $y$:

$$X = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix} \quad y = \begin{bmatrix} 1 & 0 & 1 & 0 \end{bmatrix}^T$$

Assume that $\hat{p}$ and $\hat{p}_j^{y_i}$ are estimates for $P(y = 1)$ and $P(f_j = 1 | y = y_i)$, respectively. Here, $f_i$; $i = 1, 2, 3$ is the $i^{th}$ feature. These parameters are estimated using MLE. Answer the following questions:

(i) What is the value of $\hat{p}_3^0$? Assume that no smoothing is done.

(ii) What will be the predicted label for the point $\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}^T$? Assume that no smoothing is done.

. Consider a binary classification dataset (classes are 0 and 1) with two binary features $f_1, f_2 \in 0, 1$. A naive Bayes classifier is learned and the estimated parameters are given as:

$$P(f_1 = 1|y = 0) = 0.2$$
$$P(f_2 = 1|y = 0) = 0.5$$
$$P(f_3 = 1|y = 1) = 0.6$$
$$P(f_4 = 1|y = 1) = 0.4$$

If a data point [1,0] is predicted in class 0 by this classifier, what will be the possible values for the estimate of $P(y = 1))$? Assume that tie-breaking goes to class zero. Values in the options are correct to two decimal places.

Options:

(a) $(0, 0.22]$

(b) $[0.22, 1)$

(c) $(0, 0.29]$

(d) $[0.29, 1)$

Is the following statement true or false: If $p_i^y = 0$ for $y = 0$, then $p_i^y = 1$ for $y = 1$. Here, $p_j^y$ denotes the estimate of the probability that $j^{th}$ feature value is 1 given that label is $y(P(f_i = 1|y)$.

Options:

(a) True

(b) False

7. Suppose you want to use a naive Bayes classifier to predict the gender (male or female) of a person based on two features: their height ($f_1$) and whether their is above 20 ($f_2$). Assume that the features $f_1$ and $f_2$ are conditionally independent given gender of the person, and that the variances of the height distributions $P(f_1|y = \text{male})$ and $P(f_1|y = \text{female})$ are equal. How many number of parameters are required to classify a new example using this naive Bayes classifier?

3. Consider a probability distribution over $(X, y)$ where features are one-dimensional and $y \in \{+1, -1\}$. Let $X|(y = 1)$ follow a uniform distribution over $[0, 4]$ and $X|(y = -1)$ follows a uniform distribution over $[2, 4]$

Answer the given sub-questions:

(i) If $p = P(y = 1)$ is estimated to be 0.4, what will be the prediction for the point $x = 3$ using the Bayes classifier? Enter 1 or -1.

(ii) Let $x = 2$ and let $\hat{p}$ be the estimate for $p = P(y = 1)$. Find conditions on $\hat{p}$ such that the Bayes classifier predicts 1 for this $x$. Consider that the tie-breaker is predicted in class 1.

(a) $\hat{p} \leq \frac{1}{4}$

(b) $\hat{p} \geq \frac{1}{4}$

(c) $\hat{p} \leq \frac{2}{3}$

(d) $\hat{p} \geq \frac{2}{3}$

(iii) If $p = P(y = 1)$ is estimated to be 0.5 using MLE on a given training dataset, what will be the training error of the Bayes classifier for this problem ?

. Consider a naive Bayes model is trained on the following data matrix $X$ of shape $(d, n)$ and corresponding label vector $y$:

$$X = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad y = [0, 1, 0]^T$$

Assume that $\hat{p}$ and $\hat{p}_j^{y_i}$ are estimates for $P(y = 1)$ and $P(f_j = 1 | y = y_i)$, respectively. Here, $f_i$; $i = 1, 2$ is the $i^{th}$ feature. These parameters are estimated using MLE. If a test point has label 0, what will be the probability that the point is $[0, 0]^T$?

You are working on a text classification problem using a naive Bayes classifier to determine whether an email is "Spam" or "Not Spam". You have trained your model using a dataset of 1000 emails, with 600 of them labeled as "Spam" and 400 labeled as "Not Spam". You have collected statistics on the occurrence of two words, "Won" and "Money" in these emails:

| Keyword | Label of email | Probability |
|---|---|---|
| Won | Spam | $P(Won\|\|Spam) = 0.45$ |
| Won | Not Spam | $P(Won\|\|NotSpam) = 0.05$ |
| Money | Spam | $P(Money\|\|Spam) = 0.3$ |
| Money | Not Spam | $P(Money\|\|NotSpam) = 0.02$ |

You receive a new email containing both the "Won" and "Money" keywords and want to classify it using naive Bayes. Use the naive Bayes formula to calculate the probability that the new email is classified as "Spam." (Assume that these are only two possible words, that is there are only two features).

Consider a Naive Bayes model trained on the following data matrix $X$ of shape $(d, n)$ and corresponding label vector $y$:

$$X = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \quad y = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Assume that $\hat{p}$ and $\hat{p}_j^{y_i}$ are estimates for $P(y = 1)$ and $P(f_j = 1 \mid y = y_i)$, respectively. Here, $f_i$, $i = 1, 2$, is the $i$-th feature. These parameters are estimated using Maximum Likelihood Estimation (MLE). If a test point has label 0, what will be the probability that the point is $[0, 0]^T$?

# Question

Suppose you have a five-class classification problem where class label $y \in \{0, 1, 2, 3, 4\}$ and each training example $x_i$ has binary features $f_1, f_2, f_3 \in \{0, 1\}$. How many parameters do we need to know to classify an example using Naive Bayes classifier?

# Question

Consider a probability distribution over $(X, y)$ where features are one-dimensional and $y \in \{+1, -1\}$. Let $X \mid (y = 1)$ follow a uniform distribution over $[0, 4]$ and $X \mid (y = -1)$ follows a uniform distribution over $[2, 4]$.

Let $x = 2$ and let $\hat{p}$ be the estimate for $p = P(y = 1)$. Find conditions on $\hat{p}$ such that the Bayes classifier predicts 1 for this $x$. Consider that the tie-breaker is predicted in class 1.

$Q8$) The label of every data point in a dataset can be one among $\{0, 1, 2, 3\}$. The data points belong to $\{1, 2, 3\}^5$. Find the number of parameters to be estimated if

a. Class Conditional Independence Assumption is made

b. Class Conditional Independence Assumption is not made